

The diagram illustrates the internal structure of the Raspberry Pi RP2350A QFN-60 package. Key components and connections include:

- Power Management:** A +3V3 supply is connected to the PWR\_FLAG pin. A +1V1 supply is connected to the VREG\_AVDD, VREG\_VIN, VREG\_FB, VREG\_LX, and VREG\_PGND pins. A +3V3 supply is also connected to the QSPLOVDD, IOVDD, and ADC\_AVDD pins.
- Capacitors:** C1, C2, C3, C4, C5, C6, C7, C8, C9, C10, and C11 are used for decoupling and filtering. C1 and C2 are 100nF, C3 and C4 are 27pF, C5 is 4u7, and C6, C7, C8, C9, and C10 are 4u7.
- Resistors:** R1, R2, R3, R4, R5, R6, R7, and R8 are used for various functions. R1 is 1k, R2 is 1k, R3 is 1k, R4 is 1k, R5 is 33R, R6 is 27R, R7 is 27R, and R8 is 27R.
- Inductors:** L1 and L2 are used for filtering. L1 is 3u3 and L2 is 3u3.
- Crystal:** Y1 is a 12MHz crystal.
- GPIOs:** The package includes 26 GPIO pins (GPIO0 to GPIO25) and 4 ADC pins (GPIO26 to GPIO29).
- USB:** The package includes a USB DP pin and a USB DM pin.
- RF:** The package includes an RF\_EN\_GP pin.
- Other Pins:** The package includes pins for USB\_SEL\_MCU, USB\_SEL\_CONN, TUN\_G0, TUN\_G1, TUN\_G2, TUN\_G3, ST\_LINK, ST\_INT, ST\_PWR\_EN, and LED.

The diagram shows the SPI connection between the ATmega328P and the W25Q16. The ATmega328P is represented by a yellow box with pins 1 through 7. The W25Q16 is represented by a yellow box with pins 1 through 7. The connections are as follows:

- ATmega328P Pin 1 (CS) is connected to W25Q16 Pin 1 (CS) through a 1k resistor (R6).
- ATmega328P Pin 2 (DI/IO0) is connected to W25Q16 Pin 2 (DO/IO1).
- ATmega328P Pin 3 (DO/IO1) is connected to W25Q16 Pin 3 (WP/IO2).
- ATmega328P Pin 4 (WP/IO2) is connected to W25Q16 Pin 4 (HOLD/RESET/IO3).
- ATmega328P Pin 5 (CLK) is connected to W25Q16 Pin 5 (CLK).
- ATmega328P Pin 6 (QSPI\_SCLK) is connected to W25Q16 Pin 6 (QSPI\_SCLK).
- ATmega328P Pin 7 (QSPI\_SD3) is connected to W25Q16 Pin 7 (QSPI\_SD3).

The W25Q16 is powered by +3V3 (VCC) and GND. A 100nF capacitor (C2) is connected between VCC and GND. The W25Q16 is labeled U3.

The diagram shows a switch labeled SW1 with two terminals: BOOTSEL and GND. A green wire connects the BOOTSEL terminal to the BOOTSEL pin of the microcontroller. Another green wire connects the GND terminal to a common ground point. A pull-down resistor, labeled R4 1k, is connected between the BOOTSEL pin and the common ground point.

**J2 LEFT**

|         |    |
|---------|----|
| +1V8    | 1  |
| GND     | 2  |
| TUN_G0  | 3  |
| TUN_G1  | 4  |
| CFG_SDA | 5  |
| CFG_SCL | 6  |
| GP21    | 7  |
| GP22    | 8  |
| GP23    | 9  |
| GP24    | 10 |
| GP26    | 11 |
| GP27    | 12 |
| GP28    | 13 |
| GP29    | 14 |
| +3V3    | 15 |
| VBUS    | 16 |

**J3 RIGHT**

|          |    |
|----------|----|
| ST_INT   | 1  |
| ST_LINK  | 2  |
| TUN_G2   | 3  |
| TUN_G3   | 4  |
| RF_EN_GP | 5  |
| GP7      | 6  |
| GP6      | 7  |
| GP5      | 8  |
| GP4      | 9  |
| GP3      | 10 |
| GP2      | 11 |
| GP1      | 12 |
| GP0      | 13 |
| GND      | 14 |
| +3V3     | 15 |
| VBUS     | 16 |

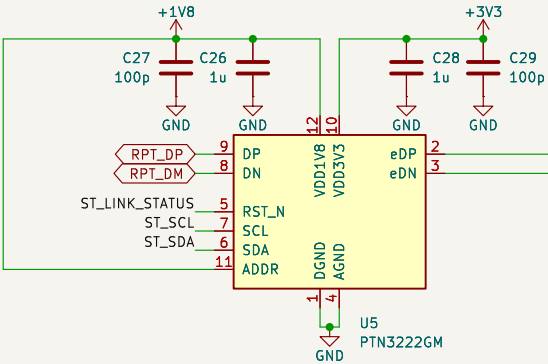
The diagram shows four red circular markers arranged horizontally, each labeled with a test point identifier and a scale value. From left to right, the labels are: H1, M2.5; H2, M2.5; H3, M2.5; and H4, M2.5.

The RP2350A, its core regulator, crystal, flash and decoupling are Raspberry Pi's RP2350A minimal design (RP-006440), part for part; the crystal is a 2520 with 27 p where that has an ABM8 and 15 p.

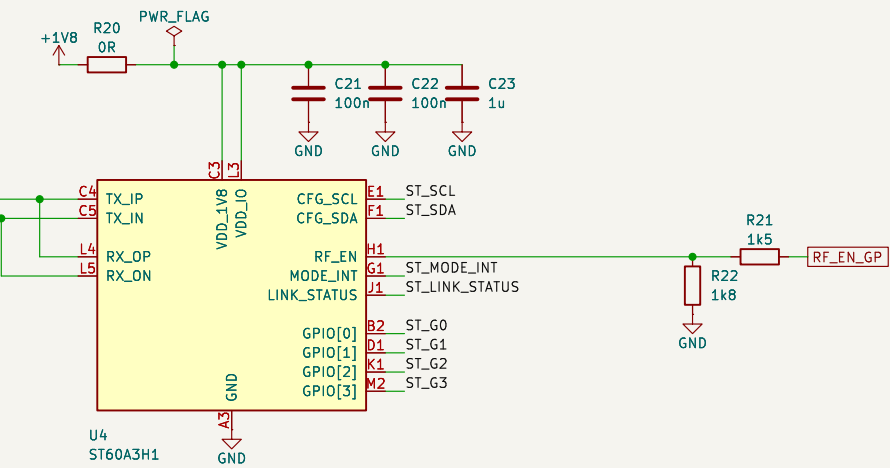
GPIO: 0-7 right-hand header. 8/9 USB path selects. 10 RF\_EN. 11 1.8V enable. 12/13 tunnelled SDA/SCL (I2C0). 14 LINK\_STATUS. 15 MODE\_SEL. 16,17 tunnelled TXD/RXD (UART0). 18/19 configuration bus (I2C1). 20 translator enable. 21-24, 26-29 left-hand header. 25 LED.

The headers carry the 3.3 V side of every line the ST60 takes from a host, so a host that is not the RP2350 can run the link.

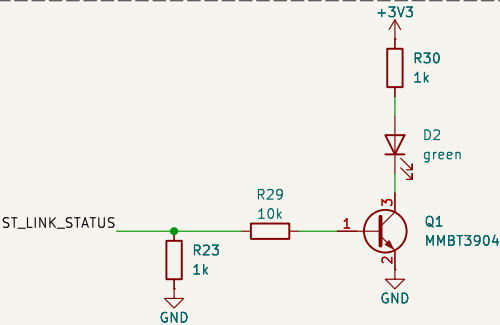
eUSB2 repeater



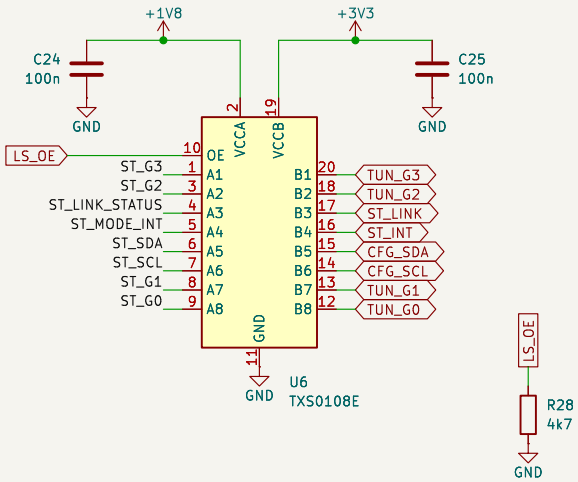
ST60A3H1



Link LED



1.8 V – 3.3 V translator



After ST's X-NUCLEO-60K1A1 (B2379A): the ST60A3H1 on +1V8 through a 0 R link, the PTN3222 on its eUSB2 pair and held in reset by LINK\_STATUS, an eight-channel auto-direction translator to 3.3 V (TXS0108E here, ST2378E there), RF\_EN by divider, the link LED off an NPN.  
The translator is off (LS\_OE low) until the RP2350 enables it, so the ST60 powers up with only its own straps on its pins. No pull-ups are fitted on the configuration bus or the tunnel pins: ST leaves its own unfitted. A tunnelled I2C bus wants pull-ups on J2/J3, at 3.3 V.  
Configuration bus: ST60A3H1 at 0x60, PTN3222 at 0x43 (ADDR on +1V8).

The schematic diagram shows the TLV75518P voltage regulator circuit. The input is +3V3, which is connected to the IN pin (pin 1) of the regulator. A 1µF capacitor (C32) is connected between the input and ground. The EN pin (pin 3) is connected to a 10k resistor (R15), which is then connected to the ST\_PWR\_EN signal. The GND pin (pin 2) is connected to ground. The output of the regulator is connected to the OUT pin (pin 5), which is then connected to the +1V8 output. A 1µF capacitor (C33) is connected between the output and ground.

|                                                                               |                  |               |
|-------------------------------------------------------------------------------|------------------|---------------|
| CERN-OHL-P                                                                    |                  |               |
| Drawn by tools/schlayout.py from the netlist in tools/circuit.py              |                  |               |
| <b>Sequoia Hope Alexander</b>                                                 |                  |               |
| Sheet: /03_usb_power/                                                         |                  |               |
| File: 03_usb_power.kicad_sch                                                  |                  |               |
| <b>Title: comms – RP2350A + ST60A3H1 60 GHz contactless link – 03_usb_pow</b> |                  |               |
| Size: A3                                                                      | Date: 2026-10-02 | <b>Rev: A</b> |
| KiCad E.D.A. 9.0.8                                                            |                  | Id: 4/4       |